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A novel framework for deep knowledge tracing via gating-controlled forgetting and learning mechanisms

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ABSTRACT

In this paper, we propose a framework called Gating-controlled Forgetting and Learning mechanisms for Deep Knowledge Tracing (GFLDKT for short). In GFLDKT, two gating-controlled mechanisms are designed to model explicitly forgetting and learning behaviors in students' learning process. With the designed gating-controlled mechanisms, both the interaction records and students' different backgrounds are combined effectively for tracing the dynamic changes of students' mastery of knowledge concepts. Results from extensive experiments demonstrate that the proposed framework outperforms the state-of-the-art models on the KT task. In addition, the ablation study shows that designed forgetting and learning mechanisms contribute clearly to the performance improvement of GFLDKT.

1. Introduction

Due to the profound impact of COVID-19 on the offline or traditional education systems, various online learning (Breslow, et al., 2013; Ramírez-Montoya, Andrade-Vargas, Rivera-Rogel, & Portuguez-Castro, 2021) platforms, such as Massive Open Online Course and Intelligent Tutoring System (Richardson, 1988), have become more and more popular in education systems. Actually, in the post-pandemic era, the convenience has made the online learning platforms as important supplements for traditional education systems (Anderson, Huttenlocher, Kleinberg, & Leskovec, 2014; Kuh, Kinzie, Buckley, Bridges, & Hayek, 2011), even as the main pattern for education systems in some extreme circumstances.

Generally, online learning systems provide a great convenience for students to learn knowledge from various materials at their own pace, which will be a main benefit for self-disciplined students or outstanding students. For average students or students without enough self-discipline, however, they are liable to get lost in the online learning systems, deriving a worse outcome than that obtained by traditional education systems. To address this issue, a feasible solution is to track the learning status of students, and to provide the corresponding intervention(s) in time to help certain students grasp required knowledge or catch up with the others in the same course.

Knowledge tracing (KT) (Corbett & Anderson, 1994), which aims to determine the students' mastery of knowledge concepts based on their historical records, is fundamental and crucial to build a successfully adaptive education system. Typically, KT utilizes machine learning-based models to complete the task of modeling the students' *knowledge state* over time. More specifically, it obtains students' proficiency status of knowledge concepts according to their interaction records (e.g. log of exercises). Given the results of KT, various applications can be conducted, such as students' performance prediction on future exercises, personalized exercises

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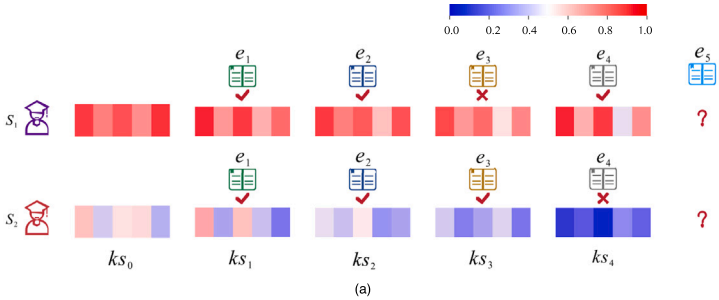

	kc_1	kc_2	kc_3	kc_4	kc_5
e_1	1	0	1	0	0
e_2	0	1	0	0	1
e_3	0	0	0	1	0
e_4	1	0	1	0	0
e_5	0	0	1	0	1

(b)

Fig. 1. A toy example to show the procedure of KT for students with different backgrounds. (a). The dynamic changes of knowledge states for two students after each interaction. Note that the color vector represents the knowledge state in certain time step for each student, and the legend of color intensity indicts the quantitative description of the mastery of certain knowledge concept. (b). The Q-matrix which describes the associated knowledge concepts for exercises. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

recommendation, and personalized learning path design. However, due to the complexity of the learning procedure (which involves psychology, pedagogy, etc.) (Forsati, Mahdavi, Shamsfard, & Sarwat, 2014; Li, Jiang, Hong, & Liao, 2017), it is not trivial to design an effective KT model. Thereafter, researchers in the literature have made extensive efforts to treat this issue according to various strategies (Cen, Koedinger, & Junker, 2006; Corbett & Anderson, 1994; Liu, et al., 2019; Nagatani, et al., 2019; Pavlik, Cen, & Koedinger, 2009; Piech, et al., 2015).

Generally, the existing approaches to KT are divided roughly into two main groups: traditional machine learning-based KT models (Corbett & Anderson, 1994) and deep learning-based KT models (Piech, et al., 2015). Bayesian Knowledge Tracing (BKT) is a classic and widely used traditional machine learning-based model for knowledge tracing, in which a Hidden Markov Model (HMM) is employed to model the knowledge concepts, and a logistic function is utilized to estimate the probability of knowledge state. Since BKT just uses skill-specific parameters (e.g., slip and guessing parameters) to model the updating of students' knowledge states, it actually models the students' learning process by only implicitly utilizing the learning curve theory, deriving the sub-optimal performance of knowledge tracing. In addition, due to the limited model capacity (Baker, et al., 2009; Pardos & Heffernan, 2011; Yudelson, Koedinger, & Gordon, 2013), BKT is not scalable to datasets consisting of exercises with many knowledge concepts. As for deep learning-based KT models, deep neural networks are applied to modeling the students' knowledge mastery by considering historical log of exercises taken by students. Thanks to the powerful representation learning ability of deep learning technologies, deep learning-based KT models are able to derive better performance by utilizing large datasets. Therefore, deep learning-based KT models are more appropriate for online learning systems, which are deriving huge data of interacting between students and the systems every day.

Until today, the representative deep learning-based KT models include Deep Knowledge Tracing (DKT) (Piech, et al., 2015), Dynamic Key-Value Memory Network (DKVMN) (Zhang, Shi, King, & Yeung, 2017), Self-attentive Knowledge Tracing (SAKT) (Pandey & Karypis, 2019), Exercise-aware Knowledge Tracing for Student Performance Prediction (EKT) (Liu, et al., 2019), Relation-aware self-attention for Knowledge Tracing (RKT) (Pandey & Srivastava, 2020), Exercise-correlated Knowledge Proficiency Tracing (EKPT) (Huang, et al., 2020), Collaborative Knowledge Tracing (CoKT) (Long, et al., 2022), etc.

Specifically, DKT applies long short-term memory (LSTM) (Hochreiter & Schmidhuber, 1997) which is the most popular recurrent neural network (RNN) to knowledge tracing. And it models the student's knowledge state with RNN's hidden vectors. Nevertheless, DKT only models implicitly the learning curve theory in process of knowledge tracing and ignores completely the forgetting curve theory. DKVMN (Zhang et al., 2017) employs the memory-augmented neural networks (MANNs) (Graves, Wayne, Reynolds, Harley, & Danihelka, 2016) to complete knowledge tracing, in which it uses static matrix keys and dynamic matrix values to store and update the students' mastery levels of knowledge concepts. However, DKVMN mainly focuses on the attention weight of exercises and knowledge concepts, while it fails to fully consider the individual differences in students' knowledge state mastery. SAKT (Pandey & Karypis, 2019) is based on the self-attention mechanism, which can better generalize on sparse data. EKT (Liu, et al., 2019) automatically learns the semantic representation of each exercise from its text content, and utilizes the attention mechanism (Vaswani, et al., 2017) to improve the predictive accuracy on students' performance. RKT (Pandey & Srivastava, 2020) introduces a relation coefficient learned from the contextual information, and then explicitly models the relation among exercises involved in past interactions for knowledge tracing. CoKT (Long, et al., 2022) utilizes the idea from collaborative filtering to combine the intra-student and inter-student information to model the update of students' knowledge states. The common drawback of the above four models (i.e., SAKT, EKT, RKT, and CoKT) lies in that they do not utilize explicitly the related theories in educational psychology, leading to the undesirable KT performance. Although EKPT (Huang, et al., 2020) utilizes learning curve and forgetting curve theories to capture the change of students' proficiency level on concepts, it fails to consider the influence of students' personalized backgrounds on the updating of knowledge states (which will be detailed by the toy example in Fig. 1 in the following paragraphs), causing the sub-optimal KT performance.

Although the above mentioned models are widely used for KT, the main drawback of most existing deep learning-based KT models lies in that some basic theories in educational psychology (e.g., distinct backgrounds or experiences among students, learning curve and forgetting curve) are not explicitly or fully employed, deriving the undesirable performance of KT. In fact, considering the

theories of learning curve and forgetting curve, the learning procedure (i.e. the update of knowledge state after interacting between students and exercises) for students with different backgrounds (e.g. outstanding students and average students) might be drastically distinct. This scenario can be demonstrated by the toy example shown in Fig. 1. According to the initial knowledge state ks_0, it is clear that the student s_1 is an outstanding student with very high grades on five knowledge concepts, while s_2 is an average student with really a poor mastery of knowledge concepts. Thereafter, the dynamic changes of knowledge states for the two students might be totally different, even though both of them have exactly the same historical interaction records.

After the first interaction in which s_2 takes the exercise e_1 and gives the correct response, the mastery status of s_2 on kc_1 and kc_3 (i.e. the two knowledge concepts involved in the exercise e_1) are improved clearly. However, since exercises e_2 and e_3 are associated with other knowledge concepts than kc_1 and kc_3, the s_2's mastery status on kc_1 and kc_3 goes down gradually, leading to the wrong response to e_4 even though it involves the same knowledge concepts as e_1. The main reason is that the theory of forgetting curve in the human learning process plays the main role for the student s_2, considering her or his poor mastery of knowledge status at the beginning. Therefore, the sensible advice for s_2 is to practice more exercises containing corresponding knowledge concepts to consolidate her or his knowledge mastery.

On the contrary, the case for student s_1 is different. From Fig. 1, we find that the mastery status on knowledge concepts for student s_1 is really stable, even though she or he gives the wrong response (probably due to a slip) to e_3. It demonstrates that outstanding students (e.g. s_1) are able to gain related knowledge even from their wrong responses. The example suggests that the student s_1 might have already grasped the corresponding knowledge concepts, and it will be better for s_1 to try exercises with higher difficulty or to learn some high-level knowledge concepts.

Based on the observations mentioned above, even having the same interaction record (i.e. taking the same exercise and giving the same response), the patterns of updating knowledge states for students might be drastically different. Therefore, we argue that an effective KT model requires to consider not only theories of learning and forgetting curves, but also the difference in backgrounds among students. In this study, we propose a novel deep learning-based KT model called Gating-controlled Forgetting and Learning mechanisms for Deep Knowledge Tracing (GFLDKT for short), in which it considers distinct roles played by theories of forgetting and learning curves on different students. More specifically, two simple but effective gating-controlled mechanisms are designed to model the learning curve and forgetting curve, respectively. Moreover, the two gating-controlled mechanisms take into account both interaction records (i.e. the exercises and the corresponding responses) and the current knowledge state (which denotes the students backgrounds). In addition, the knowledge state is associated explicitly with corresponding knowledge concepts, which makes the proposed model explainable and is more convenient to provide personalized recommendation for students in need.

In summary, the main contributions of the paper are listed as follows.

- A novel deep learning-based KT model called GFLDKT is proposed, in which the theories of learning and forgetting curves are modeled explicitly in the process of updating knowledge states.
- Two gating-controlled mechanisms are designed for learning and forgetting curves, in which the interaction records and students' distinctive backgrounds are considered simultaneously.
- Extensive experiments on four real-world datasets are conducted, and the results demonstrate the effectiveness of the proposed model.

The rest of the paper is organized as follows. Section 2 briefly reviews the related work to our study. Section 3 provides the preliminaries of the task of knowledge tracing. In Section 4, we present the details of the proposed model. In Section 5, we provide the experiments and the result analysis. Finally, we conclude this paper and provide the future directions in Section 6.

2. Related work

In this section, we review related works to the task of KT, including cognitive diagnosis which is strongly related to KT, existing KT models, and the fundamental theories of learning curve and forgetting curve in educational psychology.

2.1. Cognitive diagnosis

The procedure of cognitive diagnosis is to evaluate the knowledge state of the students through students' historical records, such as log of exercises taken by students and the corresponding responses (DiBello, Roussos, & Stout, 2006). Cognitive diagnosis models can be roughly classified into one-dimensional models and multi-dimensional models. Item Response Theory (IRT) (Reise, 2014) is a typical one-dimensional model, in which a scalar value is utilized to integrate both the students' abilities and the specific factors related to exercises (e.g. difficulty, discrimination, guessing and slipping). Specifically, the logistic regression is employed in IRT to predict students' knowledge states, based on the selected features. As for multi-dimensional models, a multi-dimensional vector is used to represent the knowledge states of students. DINA (De La Torre, 2009) is the most classic multi-dimensional model for cognitive diagnosis. In DINA, it assumes that there are K knowledge concepts of interest, and the associated knowledge concepts with exercises (which are denoted by the Q-matrix) are provided apriori by experts. Then, a K-dimensional binary vector is employed to model the mastery of knowledge concepts, each entry of which indicates the mastery status of the corresponding knowledge concept, with 1 representing mastery, and 0 otherwise. By introducing the advantage of representation learning of deep learning technology, Wang, et al. (2020) propose a model called NeuralCDM which employs deep neural networks to model complex interactions between students and exercises, deriving a better performance on cognitive diagnosis (DiBello et al., 2006).

The advantage of cognitive diagnosis models is the explainability, i.e. we can find clearly students' mastery status on knowledge concepts from the result of cognitive diagnosis models. However, the main drawback of existing models lies in that cognitive diagnosis is a static procedure which fails to model the dynamic process of knowledge tracing.

In our proposed model GFLDKT, we associate students' knowledge state explicitly with knowledge concepts, which guarantees the model's explainability. Meanwhile, the students' knowledge state is updated dynamically by considering the interaction between students and exercises in each time step. Thus, our model is able to model effectively the dynamic process of knowledge tracing.

2.2. Knowledge tracing

Knowledge tracing (KT) aims to establish a model of students' knowledge status over time through students' log of historical interactions. The first KT model is called Bayesian Knowledge Tracing (BKT) (Corbett & Anderson, 1994), in which students' mastery of knowledge is described by a latent variable in a Hidden Markov Model (HMM). Since the mastery status on each knowledge concept in BKT is a binary variable (i.e., indicating the corresponding knowledge concept has been mastered or not), it leads to the limited model capacity of BKT, which cannot be scalable to large datasets involving many knowledge concepts.

Inspired by the success of deep learning in widely applications (Abadi, et al., 2016; Devlin, Chang, Lee, & Toutanova, 2018; Duan, et al., 2017; Simonyan & Zisserman, 2014), researchers attempt to introduce deep learning into knowledge tracing (Choi, et al., 2020; Liu, Dai, Zhang, Li, & Zhang, 2020; Liu, et al., 2019; Pandey & Karypis, 2019; Piech, et al., 2015; Zhang et al., 2017). Deep Knowledge Tracing (DKT) (Piech, et al., 2015) utilizes LSTM (Hochreiter & Schmidhuber, 1997) to model students' knowledge states, achieving a better performance over traditional methods. Dynamic Key-Value Memory Network (DKVMN) (Zhang et al., 2017) uses static matrix keys to store all the knowledge concepts and a dynamic matrix values to store and update the students' mastery on knowledge concepts. By introducing the self-attention mechanism (Vaswani, et al., 2017), a transformer-based model called Self-Attentive Knowledge Tracing (SAKT) (Pandey & Karypis, 2019) considers the associations between the current exercise and exercises involved in historical interactions to improve the effectiveness of KT. Self-AttentIve Neural knowledge Tracing (SAINT) (Choi, et al., 2020), which is a variant of SAKT, tries to capture the complex relationships among different exercises and responses through deep self-attentive computations. Exercise-aware Knowledge Tracing for student performance prediction (EKT) (Liu, et al., 2019) extends the integrated state vector of each student to a knowledge state matrix, by which it can track students' knowledge states on multiple knowledge concepts simultaneously. Relation-aware self-attention for Knowledge Tracing (RKT) (Pandey & Srivastava, 2020) takes into account both the relations between exercises and the forgetting behavior to model the process of KT. Exercise-correlated Knowledge Proficiency Tracing (EKPT) (Huang, et al., 2020) attempts to model the learning process of students by using the theories of learning curve and forgetting curve, and it also employs the associations between exercises to improve the updating process of knowledge states. However, it fails to take into account the differences in the learning ability of different students, leading to the sub-optimal performance on the task of KT. By retrieving the sequences of peer students who have similar question–answering experiences, Collaborative Knowledge Tracing (CoKT) (Long, et al., 2022) obtains the inter-student information, and then integrates the inter-student information with the intra-student information to trace students' knowledge states.

The common drawback to existing KT models lies in that fundamental theories in educational psychology (e.g. learning curve (Anzanello & Fogliatto, 2011), forgetting curve (Averell & Heathcote, 2011; Ebbinghaus, 2013), and the influence of students' different backgrounds) are not explicitly utilized to model the updating of knowledge states in KT, and the explainability of KT models is limited as well.

2.3. Forgetting curve and learning curve

It is known that forgetting is inevitable in a real learning process (Markovitch & Scott, 1988). The theory of forgetting curve (Averell & Heathcote, 2011) states that the decline of memory in students' learning process usually leads to a decline in the proficiency of knowledge concepts. Ebbinghaus (Ebbinghaus, 1885) attempts to describe the forgetting curve by using a nonlinear function to associate the observed probability of memory retention with the delay or lag between the study time and the test time. Nagatani, et al. (2019) extend the deep knowledge tracing model by considering student's forgetting behavior. Specifically, the forgetting factor is modeled as an additional feature, which is integrated with the interaction embedding and previous (hidden) knowledge state to update the (new) knowledge state in KT.

Learning Curve (Anzanello & Fogliatto, 2011) specifies the improvement of knowledge as a mathematical representation in the human learning process. The learning curve theory argues that students can gain knowledge after taking each exercise. Meanwhile, the learning curve in students' learning process is affected by two factors, the properties of exercises and the experiences of students (McGrath, Guerin, Harte, Frearson, & Manville, 2015). Gan, Sun, Peng, and Sun (2020) propose a model called DKM-DLF (Knowledge Tracing Machine by modeling cognitive item Difficulty and Learning and Forgetting), in which the factorization machine framework is utilized to integrate students' abilities, the item difficulty, and the two dynamic procedures (i.e. $forgetting$ and $learning$) to model the KT.

In the proposed model GFLDKT, we model explicitly the learning curve and forgetting curve by designing two gating-controlled learning and forgetting mechanisms, respectively. Specifically, the distinct backgrounds among students are taken into account to update effectively the knowledge state in knowledge tracing.

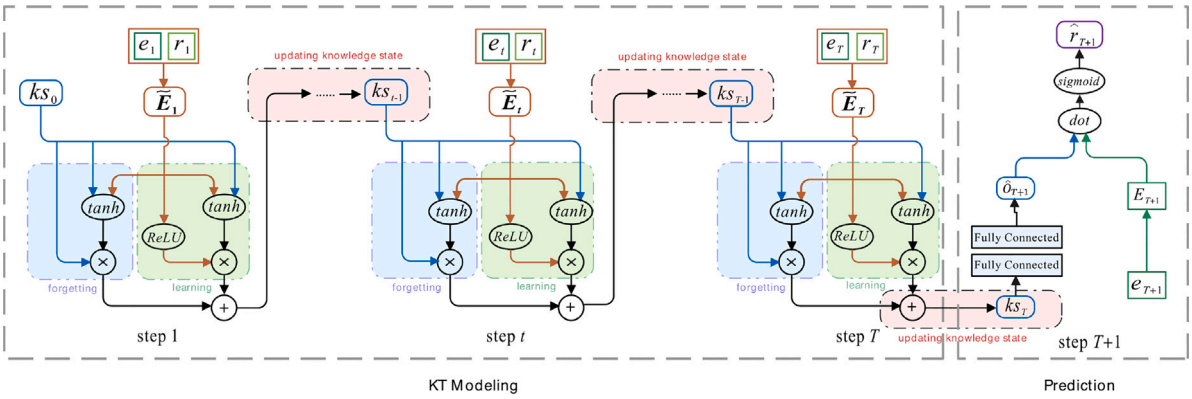

Fig. 2. The architecture of GFLDKT model.

3. Problem definition

Before presenting the detail of our model GFLDKT, in this section, we provide some preliminaries related to the task of knowledge tracing.

In this study, assume there are N students and M exercises which are associated with K knowledge concepts. The set of N students, the set of M exercises, and the set of K knowledge concepts are denoted by $SS = \{s_1, \dots, s_n, \dots, s_N\}$, $ExS = \{ex_1, \dots, ex_i, \dots, ex_M\}$, and $KS = \{k_1, \dots, k_j, \dots, k_K\}$, respectively. In addition, the associations between exercises and knowledge concepts are provided by experts, which is called Q-matrix and denoted by $\boldsymbol{Q} = \{Q_{ij}\}^{M\times K}$, where $Q_{ij} = 1$ if the exercise ex_i relates to knowledge concept k_j, and $Q_{ij} = 0$ otherwise.

For a student, the learning process in each time step is formulated as an interaction, which is denoted by (e_t, r_t) consisting of the exercise e_t taken in time step t and the response r_t provided by the student. Note that e_t denotes the index of the exercise taken in the time step t, while $r_t = 1$ means that the student provides the correct answer to the exercise e_t, otherwise $r_t = 0$. The task of knowledge tracing (KT) is to predict the response to the given exercises to a student, according to the historical records of interactions of the student. Formally, given the interaction records in previous T time step, which denoted by $R = \{(e_1, r_1), \dots, (e_t, r_t), \dots, (e_T, r_T)\}$, the KT model tries to obtain the mastery status on knowledge concepts, and derives the predicted response to the exercise (denoted by e_{T+1}) in time step $T + 1$.

The basic idea behind our model GFLDKT is that the learning and forgetting curves are modeled explicitly by using two gating-controlled mechanisms, in which the differences among students' backgrounds are considered to improve the effectiveness of updating knowledge states in each time step. The details of GFLDKT are presented in the following section.

4. GFLDKT:Gating-controlled Forgetting and Learning mechanisms for Deep Knowledge Tracing

In this section, we present the proposed framework called Gating-controlled Forgetting and Learning mechanisms for Deep Knowledge Tracing (GFLDKT for short). The framework of GFLDKT is illustrated in Fig. 2. General speaking, given the interaction set of previous T time steps $R = \{(e_1, r_1), \dots, (e_t, r_t), \dots, (e_T, r_T)\}$ for one student, GFLDKT aims to estimate/update the knowledge state of the student, according to the theories of learning curve and forgetting curve. Based on the knowledge state after the Tth time step, which is denoted by ks_T, the predicted performance (i.e. the predicted response $\hat{r}_{T+1}$) on the exercise e_{T+1} encountered in the $(T + 1)$th time step is derived by GFLDKT. The procedure of knowledge tracking in GFLDKT will be detailed in the following subsections.

4.1. Knowledge tracing modeling

Assume the initial knowledge state is denoted by kt_0 for the student. Then in the tth time step, based on the interaction record (i.e. the exercise e_t and corresponding response r_t), the current knowledge state kt_{t-1} is updated as the new knowledge state kt_t by GFLDKT. More specifically, two gating-controlled mechanisms are designed respectively for forgetting curve and learning curve. It is worth noting that through the two gating-controlled mechanisms, distinct roles played by forgetting curve and learning curve on students with different backgrounds (e.g. with different knowledge states) are modeled effectively in the process of knowledge tracing. In addition, without loss of generality, we take the tth time step as an example to describe the forgetting mechanism and learning mechanism in the following two subsections.

4.1.1. Gating-controlled Forgetting Mechanism

Forgetting is a natural process in the learning process of students. According to the observation from the example in Fig. 1, two factors will influence the forgetting mechanism in knowledge tracing, i.e. the interaction record and the mastery status on knowledge concepts (i.e. the current knowledge state). In GFLDKT, we utilize a simple but effective forgetting gate to integrate the two factors.

Assume all interactions have already been pre-trained via a pre-trained embedding module (which will be presented in Section 4.3), and the embedding (which is denoted by $\widetilde{E}_t$) of the interaction (e_t, r_t) can be obtained readily by looking up a pre-trained matrix. Moreover, the current knowledge state ks_{t-1} is a d_k-dimensional vector where d_k is a hyper-parameter of GFLDKT to control the capacity of the proposed model. It is worth noting that a transformation operator will be described in Section 4.2, by which the representation of knowledge state is transformed into the space of knowledge concepts and the explainability of GFLDKT is guaranteed.

A gating-controlled forgetting mechanism is designed to model the role of forgetting curve played on students with distinct backgrounds. Formally, the forgetting gate Γ_t^f in the tth time step is computed as follows.

$$\Gamma_t^f = tanh\left(\mathbf{W_1} \cdot ks_{t-1} + \mathbf{W_2} \cdot \widetilde{E}_t + \boldsymbol{b}^f\right) \tag{1}$$

where $tanh$ is the non-linear activation function, $\boldsymbol{W_1} \in \mathbb{R}^{d_k \times d_k}$, $\boldsymbol{W_2} \in \mathbb{R}^{d_k \times 2K}$ and $\boldsymbol{b}^f \in \mathbb{R}^{d_k}$ are the parameters of the model to be optimized.

4.1.2. Gating-controlled learning mechanism

Generally, students improve gradually their mastery of knowledge concepts through practicing step-by-step exercises involving corresponding knowledge concepts. The theory of learning curve describes the basic procedure of the improvement of knowledge concepts mastery. Based on the analysis of the toy example in Fig. 1, the same interaction (i.e. taking the same exercise and providing the same response) may induce different updates on knowledge state for different students. Therefore, an effective learning mechanism needs to take into account both the interaction and the background of students.

In GFLDKT, we utilize a learning gate Γ_t^l to model the role of learning curve played on different students. Given the current knowledge state ks_{t-1} and the interaction $\widetilde{E}_t$ in the time step t, the learning gate Γ_t^l is calculated as follows.

$$\Gamma_t^l = tanh\left(\mathbf{W_3} \cdot ks_{t-1} + \mathbf{W_4} \cdot \widetilde{E}_t + \boldsymbol{b}^l\right) \tag{2}$$

where $\mathbf{W_3} \in \mathbb{R}^{d_k \times d_k}$ and $\mathbf{W_4} \in \mathbb{R}^{d_k \times 2d_k}$ are the parameters of the proposed model. From the right hand of Eq. (2), we can find that the current knowledge state ks_{t-1} (which represents the backgrounds of different students) will influence the knowledge gained from the interaction record, through the designed learning gate.

4.1.3. Updating of knowledge state

Based on the gating-controlled forgetting and learning mechanisms, the updating of student's knowledge state in the tth time step is formally defined as follows.

$$ks_t = \Gamma_t^f \cdot ks_{t-1} + \Gamma_t^l \cdot ReLU\left(\mathbf{W_5} \cdot \widetilde{E}_t\right) \tag{3}$$

From the right hand of Eq. (3), we can find that the updating of knowledge state consists clearly of two parts. The first part is from the modeling of forgetting curve. More specifically, part of the knowledge state from the last time step contributes to the updating of the knowledge state in the current time step, in which the designed forgetting gate controls different contributions among different entries of the knowledge state.

The second part is from the knowledge gained from the interaction in the time step. The interaction embedding $\widetilde{E}_t$ is first transformed into the space of the knowledge state via a $ReLU$ function with the parameter $\mathbf{W_5}$. Then the designed learning gate is employed to consider the influence of students' backgrounds on the learning process.

4.2. Prediction module

In the $(T+1)$th time step, the current knowledge state $ks_T \in \mathbb{R}^{d_k}$ of the student is obtained via Eq. (3). For the new exercise e_{T+1}, the proposed model derives the estimated response provided by the student, according to the knowledge state ks_T. Note that the exercise representation $E_{T+1} \in \mathbb{R}^{d_k}$ can be obtained by the pre-trained exercise embedding model (which will be presented in Section 4.3.1).

To complete the prediction, two layers of fully connected networks are first utilized to map ks_T into the same space as E_{T+1}. Then the dot product of the transformed vector (which is denoted by $\hat{o}_{T+1}$) and E_{T+1} is input into a $sigmoid$ function. Finally, the predicted response is derived by a sigmoid function on the hidden representation. Formally, the process is presented as follows.

$$\hat{o}_{T+1} = ReLU\left(\mathbf{W_7} \cdot ReLU\left(\mathbf{W_6} \cdot ks_T + \mathbf{b^1}\right) + \mathbf{b^2}\right) \tag{4}$$

$$\hat{r}_{T+1} = sigmoid\left(dot(\hat{o}_{T+1}, E_{T+1})\right) \tag{5}$$

where $\mathbf{W_6}$, $\mathbf{W_7}$, $\mathbf{b^1}$ and $\mathbf{b^2}$ are parameters of the model to be trained. From the exercises embedding presented in Section 4.3.1, we can find that the representation (i.e. E_{T+1}) of each exercise has a clear association with knowledge concepts. Therefore, the

transformed representation of knowledge state, $\hat{o}_{T+1}$, will have a clear interpretation, i.e. we can illustrate the mastery status on knowledge concepts for the student. Experimental results in Section 5 demonstrate the power of the visualization of knowledge states.

Before presenting the process of training the whole framework, we introduce the pre-trained model to derive the exercise representation and the response representation in the following section.

4.3. Representation learning for exercises and responses

Based on the Q-matrix (as shown in Fig. 1), which describes the associations between exercises and knowledge concepts, a pre-trained model can be trained to use a dense vector to represent each exercise, and the corresponding interaction (i.e., each exercise and the correct or wrong response) can be obtained accordingly. The following two subsections are used to detail the exercise embedding and interaction embedding, respectively.

4.3.1. Exercise embedding

Although the Q-matrix is labeled by experts, which indicates the associations between exercises and knowledge concepts, it is not a sensible choice to use Q-matrix as the exercise embedding. There are two clear limitations for the Q-matrix: (1). it is marked manually and highly subjective, which may contain errors or subjective bias; (2). it is very sparse, which poses a great challenge for training deep learning-based models. Thereafter, in this study, we use the Q-matrix-induced partial-order-based method (Huang, et al., 2020; Rendle, Freudenthaler, Gantner, & Schmidt-Thieme, 2009) to pre-train the exercise embedding, by which each exercise is represented by a dense-valued vector. It is worth noting that each entry of the dense-valued vector (i.e. the exercise embedding) still keeps the associations between the exercise and knowledge concepts, guaranteeing the explainability of GFLDKT.

Formally, the Q-matrix is denoted by $\boldsymbol{Q} = \{Q_{ij}\}_{M\times K}$, where M and K are the number of exercises and the number of knowledge concepts, respectively. $\boldsymbol{Q_i}$ is the annotation of the ith exercise, in which $Q_{i,j} = 1$ if ith exercise relates to jth knowledge concept, and $Q_{i,j} = 0$ otherwise.

For the ith exercise, if $Q_{i,u} = 1$ and $Q_{i,v} = 0$, we define an instance of the partial-order relation, and it is denoted by $u >_i^+ v$. We argue that for the final embedding of the ith exercise, it should satisfy the partial-order constraint, which means the exercise ex_i is more related to the uth knowledge concept than the vth knowledge concept. Given the Q-matrix, the induced partial-order set is defined as a set of three-tuples, which is denoted by $\boldsymbol{P_Q}$. Formally, $\boldsymbol{P_Q}$ is defined as follows.

$$\boldsymbol{P_Q} = \{(i,u,v) \mid u >_i^+ v, \text{ for } i \in \{1,\dots,M\}\} \tag{6}$$

Generally, the exercise embedding method used in this study aims to pre-train an embedding matrix $\boldsymbol{A} \in \mathbb{R}^{M\times K}$ which satisfies the Q-matrix induced partial order set $\boldsymbol{P_Q}$ as much as possible. Specifically, a generative model is designed for the exercise embedding, and a Bayesian treatment is utilized to complete the optimizing process. Formally, the posterior probability of $\boldsymbol{P_Q}$ given the embedding matrix $\boldsymbol{A}$ is computed as follows.

$$p(\boldsymbol{A} \mid \boldsymbol{P_Q}) \propto p(\boldsymbol{P_Q} \mid \boldsymbol{A}) \times p(\boldsymbol{A}) \tag{7}$$

where $p(\boldsymbol{P_Q} \mid \boldsymbol{A})$ and $p(\boldsymbol{A})$ are likelihood distribution and prior distribution, respectively.

For the prior distribution, we assume it follows a zero-mean spherical Gaussian distribution with the scalar hyper-parameter $\sigma_{\boldsymbol{A}}^2$. Specifically, for each exercise ex_i, the embedding vector is generated as follows in Eq. (8).

$$p(\boldsymbol{A}_i) \sim \mathcal{N}\left(0, \sigma_{\boldsymbol{A}}^2 \mathbf{I}\right) \tag{8}$$

Thereafter, the prior distribution for the whole embedding matrix $p(\boldsymbol{A})$ is computed as follows.

$$\begin{aligned} p(\boldsymbol{A}) &= \prod_{i=1}^{M} \mathcal{N}\left(\boldsymbol{A}_i; 0, \sigma_{\boldsymbol{A}}^2\right) \\ &= \prod_{i=1}^{M}\prod_{k=1}^{K} \frac{1}{\sqrt{2\pi}\sigma_{\boldsymbol{A}}} e^{-\frac{\boldsymbol{A}_{i,k}{}^2}{2\sigma_{\boldsymbol{A}}^2}\mathbf{I}} \end{aligned} \tag{9}$$

For the likelihood distribution, we assume the partial orders induced by Q-matrix are I.I.D. among exercises. Formally, the likelihood distribution is defined as follows.

$$p(\boldsymbol{P_Q} \mid \boldsymbol{A}) = \prod_{i=1}^{M} \prod_{(i,u,v)\in\boldsymbol{A}_i} p(u >_i^+ v \mid \boldsymbol{A}_i) \tag{10}$$

Given each exercise ex_i, in order to make the exercise embedding $\boldsymbol{A}_i$ satisfy the induced partial order constraints, the likelihood distribution is defined by a *sigmoid* function as follows.

$$p(u >_i^+ v \mid \boldsymbol{A}_i) = \sigma(\boldsymbol{A}_{i,u} - \boldsymbol{A}_{i,v}) \tag{11}$$

where $\sigma(x)$ is the *sigmoid*, i.e. $\sigma(x) = \frac{1}{1+e^{-x}}$.

Finally, by combining Eq. (9), (10) and (11), the log posterior of the embedding matrix $\boldsymbol{A}$ is derived as follows.

$$\begin{aligned}
&\ln p(\boldsymbol{A} \mid \boldsymbol{P_Q}) \\
&= \ln p\left(\boldsymbol{P_Q} \mid \boldsymbol{A}\right) + \ln p(\boldsymbol{A}) \\
&= \ln \prod_{(i,u,v)\in \boldsymbol{P_Q}} p\left(u >_i^+ v \mid \boldsymbol{A}\right) + \ln \prod_{i=1}^{M} \mathcal{N}\left(\boldsymbol{A}_i; 0, \sigma_{\boldsymbol{A}}^2 \mathbf{I}\right) \\
&= \sum_{i=1}^{M}\sum_{u=1}^{K}\sum_{v=1}^{K} I(u >_i^+ v \in \boldsymbol{P_Q}) \ln \frac{1}{1+e^{-(\boldsymbol{A}_{i,u}-\boldsymbol{A}_{i,v})}} \\
&- \sum_{i=1}^{M}\sum_{k=1}^{K} \frac{\boldsymbol{A}_{i,k}^2}{2\sigma_{\boldsymbol{A}}^2} + \text{Const.}
\end{aligned} \tag{12}$$

where $I(\cdot)$ denotes an indication function which derives 1 if the assertion inside is true and derives 0 otherwise. In addition, Const. denotes the constant that can be ignored during optimization.

The gradient of log posterior with respect to each model parameter $\boldsymbol{A}_{i,k}$ ($i \in \{1, \dots, M\}$, $k \in \{1, \dots, K\}$) is derived as follows.

$$\begin{aligned}
&\frac{\partial \ln p(\boldsymbol{A} \mid \boldsymbol{P_Q})}{\partial \boldsymbol{A}_{i,k}} = \\
&\sum_{v=1}^{K} I(k >_i^+ v \in \boldsymbol{P_Q}) \cdot (1+e^{-(\boldsymbol{A}_{i,k}-\boldsymbol{A}_{i,v})} \cdot e^{-(\boldsymbol{A}_{i,k}-\boldsymbol{A}_{i,v})}) \cdot (-1) \\
&+ \sum_{u=1}^{K} I(u >_i^+ k \in \boldsymbol{P_Q}) \cdot (1+e^{-(\boldsymbol{A}_{i,u}-\boldsymbol{A}_{i,k})} \cdot e^{-(\boldsymbol{A}_{i,u}-\boldsymbol{A}_{i,k})}) \cdot (1) \\
&- \frac{\boldsymbol{A}_{ik}}{\sigma_{\boldsymbol{A}}^2}
\end{aligned} \tag{13}$$

Note that the first part of the *r.h.s* of Eq. (13) is the contribution from the partial orders where kth knowledge concept appears as the first element, while the second part is the contribution from the partial orders where kth knowledge concept appears as the second element. Since the coupling of parameters in Eq. (13), we cannot derive a closed form for updating the parameter $\boldsymbol{A}_{i,k}$. Instead, the iterative update of $\boldsymbol{A}_{i,k}$ is used as follows.

$$\boldsymbol{A}_{i,k}^{new} = \boldsymbol{A}_{i,k}^{old} + \rho \cdot \frac{\partial \ln p(\boldsymbol{A}^{old} \mid \boldsymbol{P_Q})}{\partial \boldsymbol{A}_{i,k}} \tag{14}$$

where ρ is the learning rate.

Through applying Eq. (14) in an iterative manner, the embedding matrix $\boldsymbol{A}$ can be optimized accordingly. And $\boldsymbol{A} \in \mathbb{R}^{\boldsymbol{M}\times\boldsymbol{K}}$ can be used as a look-up table in the process of knowledge tracing. For example, in the tth time step, the exercise taken by a student is denoted by e_t, which is a one-hot vector, representing the index of the exercise. In other words, if the mth entry of e_t is 1 (of course the other entries are 0), the student takes the mth exercise in the whole set of exercises in the tth time step. The embedding of the exercise e_t is denoted by E_t which is obtained by Eq. (15).

$$E_t = \boldsymbol{A}^{\top} \times e_t \tag{15}$$

4.3.2. Interaction embedding

Given the interaction record (e_t, r_t) in the tth time step, the embedding (denoted by $\widetilde{E}_t$) is a $2K$-dimensional vector, which is defined formally as follows.

$$\widetilde{E}_t = \begin{cases} \left[E_t \oplus \mathbf{0}\right], & r_t = 1 \\ \left[\mathbf{0} \oplus E_t\right], & r_t = 0 \end{cases} \tag{16}$$

where $\mathbf{0}$ is a K-dimensional vector with all 0 entries and E_t is derived via Eq. (15). It is clear that two interactions consisting the same exercise but different responses (correct vs. wrong) will have very different embeddings.

This section describes the pre-training model for exercise embedding and interaction embedding, which can be used as lookup tables for knowledge tracing module (as presented in Section 4.4) and prediction module (as presented in Section 4.2).

4.4. Model training

In order to explain our idea more clearly, in this subsection, we describe the training process of the proposed model, i.e. how to determine the parameters of GFLDKT based on the training data.

Assume the size of batch/mini-batch is N, i.e. there are interaction records of N students in each training batch/mini-batch. Assume also that the interaction record of each student consists of interactions in T time steps. Without loss of generality, for student n, the list of interactions is denoted as $((e_{n,1}, r_{n,1}), \dots, (e_{n,t}, r_{n,t}), \dots, (e_{n,T}, r_{n,T}))$. In each time step t, the current knowledge state ks_{t-1} is first calculated by the KT modeling (which is presented in Section 4.1). Then the predicted response $\widehat{r}_{n,t}$ is derived via Eqs. (4)

Table 1
The statistics of datasets.

	Junyi	Assist15	Assist17	Static11
#Students	238,120	19,840	1,709	333
#Knowledge concepts	828	100	102	1,223
#Responses	26,666,117	683,801	942,816	189,297
#Responses per student	120	34	552	568

and (5) (which is presented in Section 4.2). By comparing the difference between $\hat{r}_{n,t}$ and the ground-truth response $r_{n,t}$, a loss in time step t can be computed.

In this paper, the cross-entropy loss function is utilized to optimize the proposed model. For each batch/mini-batch, the total loss is defined formally as follows.

$$\mathcal{L} = -\sum_{n=1}^{N}\sum_{t=1}^{T}(r_{n,t}\log\hat{r}_{n,t} + (1 - r_{n,t})\log(1 - \hat{r}_{n,t})) \tag{17}$$

Through minimizing the loss function in Eq. (17) via using the optimization algorithm Adam (Kingma & Ba, 2015), the optimized GFLDKT is obtained, which is used for prediction accordingly. More details of experimental settings will be provided in the following section.

5. Experiments

In this section, we present comprehensive experiments to evaluate the proposed framework from four aspects, including the performance comparison of GFLDKT with several selected baselines, the evaluation of contributions of designed $forgetting$ and $learning$ mechanisms, a case study to demonstrate the visualization of knowledge states in knowledge tracing, and the hyper-parameter sensitivity analysis.

5.1. Dataset description

In our experiments, the following four public datasets are used for predictive performance evaluation.

- **JunyiAcademy(Junyi)**[1] This dataset (Chang, Hsu, & Chen, 2015) was collected by JunyiAcademy[2], which focuses on basic education assistance for primary and secondary schools, covering learning resources from main subjects to computer sciences, humanities and test evaluation. The Junyi dataset consists of 26,666,117 responses provided by 238,120 students on 828 knowledge concepts.
- **ASSISTments2015(Assist15)**[3] This dataset was collected from ASSISTments in 2015. The Assist15 dataset contains 683,801 responses by19,840 students on 100 knowledge concepts.
- **ASSISTments2017(Assist17)**[4] This dataset was provided by ASSISTments Data Mining Competition 2017. The Assist17 dataset consists of 942,816 responses by 1,709 students on 102 knowledge concepts.
- **Statics2011(Static11)**[5] The Static11 dataset was collected from a statistics course at Carnegie Mellon University during Fall 2011, which consists of 189,297 responses by 333 students on 1,223 knowledge concepts.

The detailed statistics of all datasets are summarized in Table 1.

5.2. Experimental setup

In this subsection, the detailed experimental setup is provided, including the implementation details of the proposed model, selected baselines, and evaluation metrics.

[1] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=1275
[2] https://www.junyiacademy.org/
[3] https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data
[4] https://sites.google.com/view/assistmentsdatamining/dataset
[5] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507

5.2.1. Implementation details

In our implementation of GFLDKT, the dimension d_k of knowledge state and the dimension of exercise embedding d_M are set as 50 and 25, respectively. The dimensions of hidden representations derived by the two full connection layers (as shown in Eq. (4)) are 512 and 256, respectively. The hyperparameter σ_A in "Exercise Embedding" (as shown in Eq. (8)) is set as 0.1. To optimize our model, the learning rate and dropout rate are set as 0.01 and 0.3, respectively. Note that the settings of three hyper-parameters (i.e. d_k, σ_A^2, and dropout rate) are determined by a study of sensitivity analysis which will be presented in Section 5.3.4.

In experiments, we filter out students with less than 30 responses in each dataset to guarantee that each student has enough historical interactions for knowledge tracing. For each dataset, we conduct 5-fold cross-validation for all KT models, in which each dataset is split according to students. The parameters deriving the best performance in 5-fold cross-validation were used for performance evaluation.

In addition, the trend of predictive performance on different sizes of historical interactions is investigated for our model and selected baselines. Specifically, for each student, the different sizes of beginning parts (about 60%, 70%, 80%, 90% of whole interactions) of exercising records are utilized for model training, and the remains are used for testing.

5.2.2. Baselines and evaluation metrics

In order to demonstrate the effectiveness of the proposed framework, the following models are selected as baselines in our experiments.

- BKT: Bayesian Knowledge Tracing (BKT) (Corbett & Anderson, 1994) model represents a student's knowledge state as a binary variable for each knowledge concept. BKT builds a separate HMM model for each knowledge concept to predict students' mastery of a specific knowledge concept.
- IRT: Item Response Theory (IRT) (DiBello et al., 2006) model uses a scalar value to represent students' knowledge state and employs a set of parameters related to the item to analyze the probability of correctly answering the exercises.
- DKT: Deep Knowledge Tracing (DKT) (Piech, et al., 2015) uses the LSTM (Hochreiter & Schmidhuber, 1997)to capture the latent knowledge states and its temporal dynamics.
- $SAKT$: Self-Attentive model for Knowledge Tracing (SAKT) (Pandey & Karypis, 2019) utilizes self-attention to identify the relevance between KCs, and predicts the student's performance on the new exercise by considering the different associations between the exercise and previously answered exercises.
- $DKVMN$: Dynamic Key-Value Memory Networks (DKVMN) (Zhang et al., 2017) introduces the Key-Value Memory Networks to model students' historical interactions, by which the performance on new exercises is predicted.
- $DKT + Forgetting$: This model (Nagatani, et al., 2019) extends the DKT mode (Piech, et al., 2015) by incorporating information related to the forgetting curve for knowledge tracing.
- RKT: Relation-Aware Self-Attention for Knowledge Tracing(RKT) (Pandey & Srivastava, 2020) introduces a relation-aware self-attention layer to model the hidden relations between exercises.
- EKT: Exercise-aware Knowledge Tracing (EKT) (Liu, et al., 2019) uses a bidirectional LSTM to encode exercise text, by which a better embedding of exercises is obtained and the prediction performance is improved accordingly.
- $EKPT$: Exercise-correlated Knowledge Proficiency Tracing (EKPT) considers theories of learning curve and forgetting curve to capture the updating of students' knowledge states, and incorporates the connectivity between concepts contained in exercises into the process of knowledge tracing.
- $CoKT$: Collaborative Knowledge Tracing (CoKT) (Long, et al., 2022) integrates the inter-student information with the intra-student information to trace students' knowledge states and predict the responses to the new exercises.

Note that for each baseline model, the parameters or hyper-parameters were set by the method introduced in the original paper.

To compare the performance of the proposed model and selected baselines, the Area Under the Curve (AUC) and Accuracy (ACC) are used as evaluation metrics in experiments.

5.3. Experimental results

5.3.1. Predictive performance comparison

Table 2 shows the prediction results of our model and selected baselines according to AUC and ACC. Generally speaking, the proposed model GFLDKT outperforms baselines on all of the four datasets.

BKT and IRT derive the worst performance among all models, indicating that traditional machine learning-based models have the limited capacity and fail to learn the complex semantic associations among large datasets. DKT and DKVMN provide better results than traditional machine learning-based models (i.e. BKT and IRT), suggesting the powerful ability of representation learning in deep learning-based models. However, both models still derive worse performance than those derived by other deep learning-based KT models. The main reason is that these two models use relatively simple deep learning models which are not powerful enough to capture the hidden information contained in students' historical interactions. The interesting observation is that DKT+Forgetting derives better performance than DKT, which demonstrates that explicitly modeling the forgetting curve can improve the effectiveness of KT models. In addition, the other three baselines (i.e. SAKT, RKT and EKT), which employ attention mechanisms to model relatively finer-grained information in training data, derive better performance than DKT and DKVMN. More specifically, SAKT is a KT model implemented on Transformers, and the self-attention layer is used to capture the relevance among KCs. RKT and EKT use

Table 2
Comparison results of predictive performance.

Methods	Junyi		Assist15		Assist17		Static11	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
BKT	0.623	0.597	0.645	0.613	0.631	0.622	0.637	0.621
IRT	0.714	0.611	0.702	0.688	0.697	0.657	0.721	0.720
DKT	0.767	0.623	0.723	0.701	0.711	0.691	0.749	0.756
SAKT	0.819	0.728	0.703	0.736	0.706	0.702	0.803	0.766
DKVMN	0.814	0.714	0.727	0.734	0.734	0.711	0.777	0.784
DKT+Forgetting	0.838	0.732	0.731	0.727	0.73	0.708	0.769	0.779
RKT	0.857	0.753	0.745	0.751	0.736	0.747	0.819	0.792
EKT	0.847	0.755	0.753	0.749	0.751	0.733	0.826	0.803
EKPT	0.851	0.759	0.755	0.751	0.761	0.749	0.827	0.811
CoKT	0.856	0.762	0.759	0.753	0.764	0.751	0.829	0.807
GFLDKT	**0.861**	**0.773**	**0.769**	**0.756**	**0.782**	**0.762**	**0.832**	**0.814**

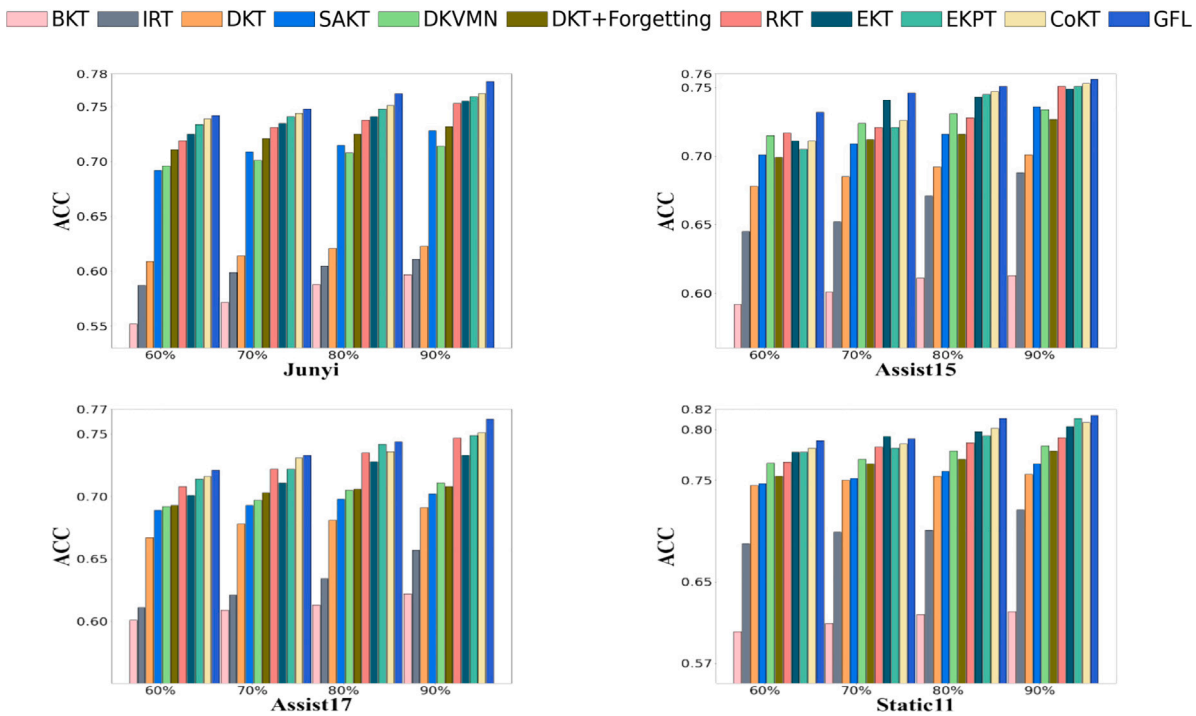

Fig. 3. Results on trend of predictive performance with different sizes of historical interactions according to ACC.

attention mechanisms to model the relations among exercises by which the associations among KCs are captured implicitly, deriving better predictive performance. As for EKPT, although it also uses the theories of learning curve and forgetting curve, it ignores the different backgrounds of students, leading to the sub-optimal performance according to both AUC and ACC. CoKT, which is the SOTA model for KT, gives the best results among selected baselines, showing that combining the intra-student and inter-student information can learn more accurate knowledge states for students.

As for the proposed model GFLDKT, the forgetting and learning mechanisms are modeled explicitly for students with different backgrounds, by which the dynamic changes of knowledge state mastery are captured effectively from the historical interactions. Therefore, GFLDKT outperforms selected baselines on the datasets by a clear margin according to the evaluation measurements (i.e. AUC and ACC).

In addition, Figs. 3 and 4 present the comparison results on the trend of prediction performance of KT models with different sizes of historical interactions in training data according to both ACC and AUC metrics. From Figs. 3 and 4, two observations are obtained: (1). all models derive better results with more historical interactions (i.e. records on more exercises) in training data; (2). our model GFLDKT performs stably better than baselines.

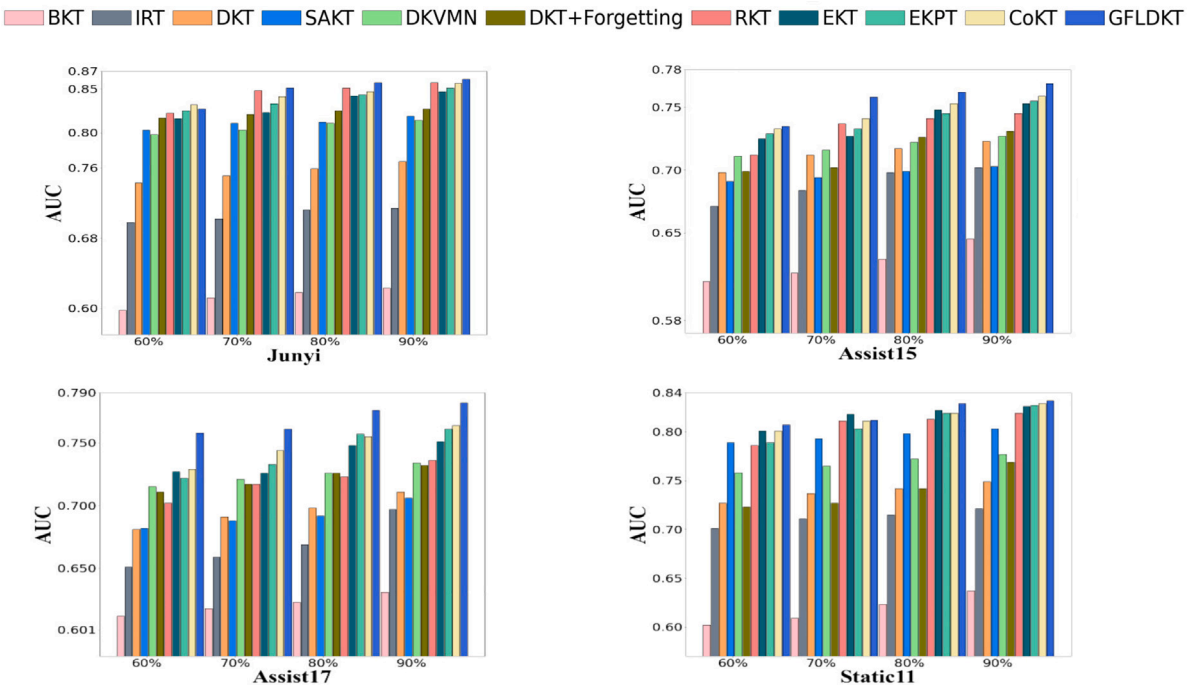

Fig. 4. Results on trend of predictive performance with different sizes of historical interactions according to AUC.

Table 3
Results of ablation study.

Methods	Junyi		Assist15		Assist17		Static11	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
GFLDKT w/o Forgetting	0.816	0.722	0.745	0.725	0.751	0.677	0.801	0.787
GFLDKT w/o Learning	0.843	0.716	0.727	0.736	0.761	0.698	0.799	0.769
GFLDKT	**0.861**	**0.773**	**0.769**	**0.756**	**0.782**	**0.762**	**0.832**	**0.814**

The results in Table 2, Figs. 3 and 4 suggest that explicitly modeling the forgetting and learning mechanisms plays important roles in accurately capturing students' knowledge states in the learning process. The detailed contributions of modeling forgetting and learning mechanisms to improving predictive performance will be presented in the following subsection.

5.3.2. Effectiveness of modeling forgetting and learning mechanisms

In this section, a thorough ablation study is conducted to evaluate the effectiveness of the main components in GFLDKT, and the results are shown in Table 3. Note that "GFLDKT w/o Forgetting" and "GFLDKT w/o Learning" denote the two variants of GFLDKT, respectively. Specifically, "GFLDKT w/o Forgetting" denotes GFLDKT without modeling the distinctive forgetting mechanism for students with different backgrounds (e.g. outstanding students and average students). In other words, Eq. (1) is replaced by $\Gamma_t^f = tanh\left(\mathbf{W_2} \cdot \widetilde{E}_t + \boldsymbol{b}^f\right)$ for the variant model "GFLDKT w/o Forgetting", in which the forgetting mechanism is only determined by students' interactions. The variant "GFLDKT w/o Learning" denotes GFLDKT without modeling the distinctive learning mechanism for different students, i.e. Eq. (2) is replaced by $\Gamma_t^l = tanh\left(\mathbf{W_4} \cdot \widetilde{E}_t + \boldsymbol{b}^l\right)$ for "GFLDKT w/o Learning", in which the learning mechanism is only determined by students' interactions.

The results in Table 3 demonstrate that effectively modeling personalized forgetting and learning patterns is of importance for the performance improvement of the proposed model GFLDKT. More specifically, the modeling of distinctive forgetting patterns for students with different backgrounds plays a vital role in improving the performance of the model. If we fail to take distinctive forgetting patterns into account when modeling students' knowledge states, the predictive performance of our model will decrease significantly, by comparing with the results derived by GFLDKT and the variant "GFLDKT w/o Forgetting". Moreover, the designed gating-controlled learning mechanism is also important for the effectiveness of GFLDKT. In a word, the results of ablation study demonstrate the effectiveness of the designed forgetting and learning mechanisms in GFLDKT, and provide as well the detailed explanation for the best results derived by GFLDKT in Table 2, Figs. 3 and 4.

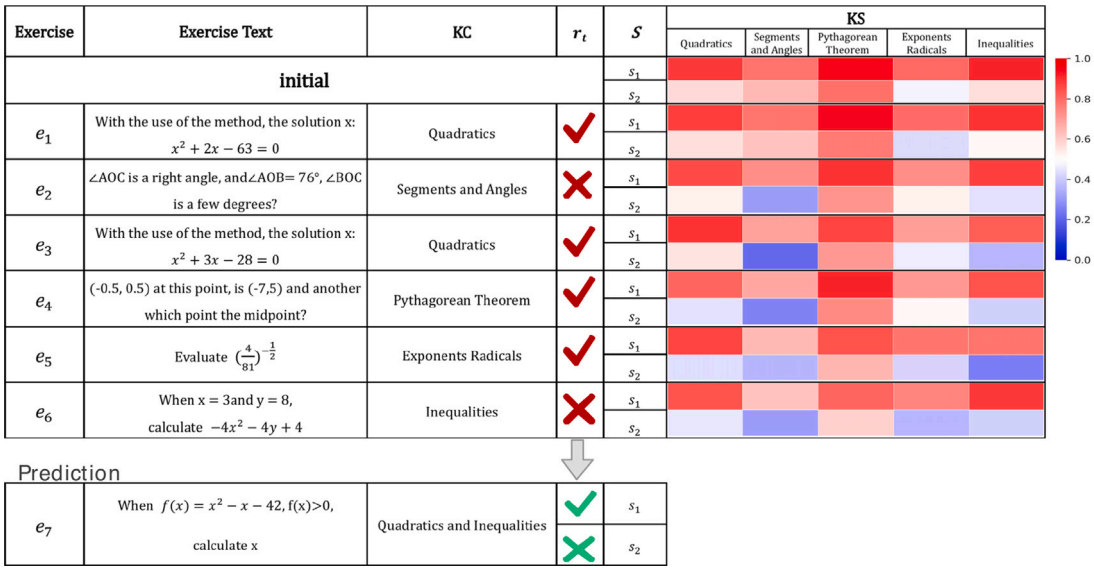

Fig. 5. The illustration of the case study on Junyi for two students. The predictive performance on e_7 is derived based on the past six interactions. In the right part, the bar plots show the knowledge state (KS) on corresponding knowledge concepts (KC) of the two students after each interaction.

5.3.3. Case study

In this section, we present a case study for GFLDKT on the Junyi dataset for two students (denoted by s_1 and s_2, respectively), and the results are presented in Fig. 5. It is clear that s_1 is an outstanding student with a better initial mastery of knowledge concepts before the interactions while s_2 is an average student with a relatively ordinary basis of knowledge states.

From Fig. 5, we can find that even though the responses of the two students are exactly the same in previous six interactions, dynamic changes of knowledge states derived by GFLDKT are drastically different. For example, although both s_1 and s_2 provide wrong answers to the exercise e_2 (which is associated with the KC "Segments and Angles"), the knowledge state on the KC "Segments and Angles" of s_1 does not change significantly while the mastery of the KC "Segments and Angles" of s_2 decreases clearly. The main reason is that for outstanding students (e.g. s_1), they can even learn knowledge concepts from errors they made. However, for the average students, the negative responses suggest that they need to improve their mastery on certain knowledge concepts. For the forgetting behavior, we can find the similar pattern with the learning behavior for students with different backgrounds. Before predicting the response on the exercise e_7 (which is associated with the KC "Quadratics and Inequalities"), s_2's mastery on "Quadratics" goes down clearly since the last exercise involving "Quadratics" is in three time steps ago (i.e. the exercise e_3), for which the forgetting mechanism plays the main role in the mastery change of KCs. However, for s_1 who is an outstanding student, the forgetting mechanism does not affect the mastery status of certain KCs.

According to the previous interactions, GFLDKT derives the positive and negative predicted responses respectively on the exercise e_7 for s_1 and s_2, both of which are accurate predictions by manually checking the ground-truth annotations. The results of the case study demonstrate again the effectiveness of modeling explicitly the personalized forgetting and learning patterns in the proposed model GFLDKT.

5.3.4. Hyper-parameter sensitivity analysis

In this section, we study the impact of hyper-parameters of the proposed model according to AUC on the Junyi dataset. More specifically, we present the sensitivity analysis of three important hyper-parameters in GFLDKT, including σ_A^2 in the process of exercise embedding (as shown in Eq. (8)), the dimension of knowledge state d_k, and the dropout rate. The results are presented in Fig. 6. Note that each sub-figure demonstrates the results obtained by GFLDKT with a fixed value of σ_A^2 (i.e. $\sigma_A^2 \in \{0.01, 0.1, 1, 10\}$).

- σ_A^2. For the variance of the prior distribution of exercise embedding, we test four settings (i.e. $\sigma_A^2 \in \{0.01, 0.1, 1, 10\}$) for the sensitivity analysis. From results in Fig. 6, we observe the similar trend of predictive performance for different values of σ_A^2. And GFLDKT with $\sigma_A^2 = 0.1$ derives the best performance with 0.861 for AUC (with $d_k = 50$ and $dropout\ rate = 0.3$). Thereafter, for performance comparison with selected baselines, we use the results derived by GFLDKT with $\sigma_A^2 = 0.1$ in previous experiments.
- d_k. For dimension of the knowledge state, the results show the similar patterns for GFLDKT with different settings of σ_A^2. As the dimension of d_k increases from 25 to 50, the predictive performance goes up. After that, the performance begins to go gradually down. The results indicate that a vector with a bigger dimension can generally encode more meaningful information for knowledge states, but a vector with too big size may introduce additional redundancies and lead to over-fitting. From the results in Fig. 6, we find that the setting of d_k as 50 can derive the best experimental results.

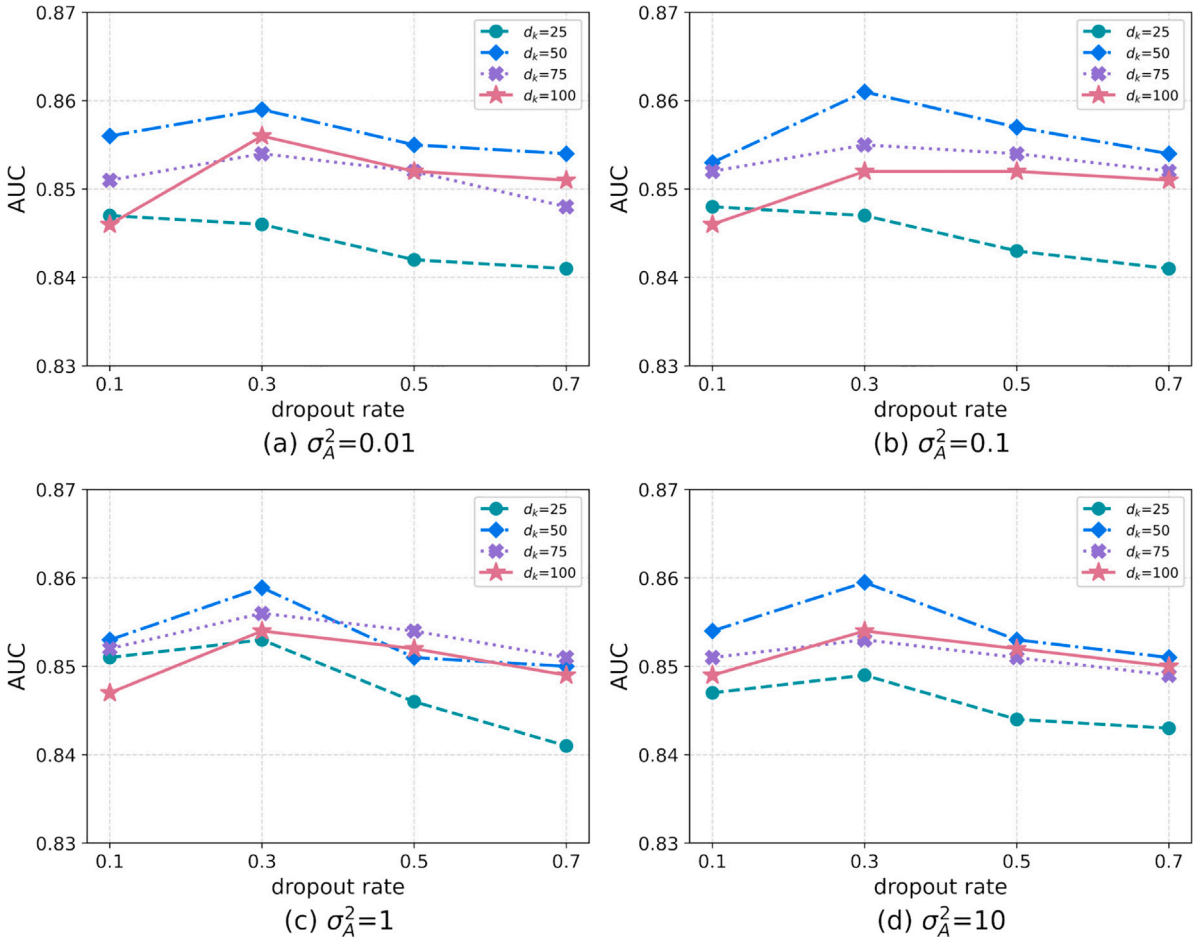

Fig. 6. Results of hyper-parameter sensitivity analysis.

- *Dropout Rate*. We test four different values of dropout rate (i.e. *dropout rate* $\in \{0.1, 0.3, 0.5, 0.7\}$) in this study. With changing dropout rate from 0.1 to 0.3, the predictive performance is improved accordingly, and the best AUC 0.861 is obtained when setting $\sigma_A^2 = 0.1$ and $d_k = 50$. However, when setting a bigger dropout rate (i.e., 0.5, 0.7), the performance of the proposed method will go down gradually. Therefore, we set dropout rate as 0.3 in the implementation of GFLDKT for performance comparison with baselines.

6. Conclusions and future work

In this paper, we have proposed a novel knowledge tracing model called GFLDKT, in which forgetting and learning theories in educational psychology are considered explicitly for modeling students' mastery of knowledge concepts. More specifically, two gating-controlled mechanisms are designed for forgetting and learning curves, in which the interaction records and students' backgrounds are considered simultaneously. With the designed gating-controlled mechanisms, the influence of the students' backgrounds on the forgetting and learning behaviors is effectively captured, and the accurate mastery of knowledge states is obtained accordingly. Extensive experiments on four publicly available datasets have been conducted. The results of predictive performance comparison show that GFLDKT outperforms existing representative models on the task of knowledge tracing. The results of ablation study demonstrate that considering students' different backgrounds in the modeling process of forgetting and learning mechanisms is important for the effectiveness of the proposed model. In addition, the results of the case study suggest that GFLDKT provides the explainability in the updating process of students' knowledge states.

In the future, we plan to explore our model in other online educational applications, such as personalized question recommendation, and personalized learning path recommendation.

CRediT authorship contribution statement

Weizhong Zhao: Conceptualization, Methodology, Investigation, Writing – original draft. **Jun Xia:** Investigation, Writing – original draft. **Xingpeng Jiang:** Validation, Investigation. **Tingting He:** Visualization, Writing – review & editing.

Data availability

Data will be made available on request.

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